Lecture 4: Linear Independence and Dimension

Proving Rank is Well-Defined

Motivation: Is Rank Well-Defined?

Recall: Rank from Cross-Filling

In Lecture 3, we defined:

$$\text{rank}(A) = \text{number of rank-one pieces from cross-filling}$$

Problem: Cross-filling allows **different pivot choices**.

For

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$$

we could pick pivot at $(1, 1)$, or $(2, 2)$, or $(3, 3)$.

Question: Do all choices give the **same number** of pieces?

Today's Goal

Prove that rank is well-defined by showing:

$$\boxed{\text{rank}(A) = \text{dimension of column space}}$$

Strategy:

1. Introduce **linear independence** and **basis**
2. Prove all bases have the **same size** (using trace)
3. Show cross-filling produces a **basis** for column space
4. Conclude: rank = dimension (well-defined!)

Linear Independence

Linear Independence — Definition

Definition (Linear Independence)
Vectors $v_1, v_2, \ldots, v_k$ are **linearly independent** if:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0 \quad \Rightarrow \quad c_1 = c_2 = \cdots = c_k = 0$$

In other words: The **only** way to make zero is the **trivial combination**.

Concrete Example: Coffee Shop Recipe Vectors

Shinchan's coffee shop has 3 recipe vectors (columns = ingredients):

	☕	🍵	🥛
🫘	2	1	0
🍃	1	2	0
🍋	0	1	0
🐄	1	0	1

These are recipe vectors $v_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix}$, $v_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

Question: Are these linearly independent?

Testing Independence: Can We Make Zero Recipe?

We want to test if $c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ forces $c_1 = c_2 = c_3 = 0$.

Think about it physically: Can we combine recipes to get **zero of all ingredients**?

$c_1 \cdot$ ☕ $+ c_2 \cdot$ 🍵 $+ c_3 \cdot$ 🥛 $=$ nothing

Ingredient	Amount
🫘	$2c_1 + 1c_2 + 0c_3 = 0$
🍃	$1c_1 + 2c_2 + 0c_3 = 0$
🍋	$0c_1 + 1c_2 + 0c_3 = 0$
🐄	$1c_1 + 0c_2 + 1c_3 = 0$

Solution: Backward Substitution

From the equations:

$$\begin{aligned} 2c_1 + c_2 &= 0 \\ c_1 + 2c_2 &= 0 \\ c_2 &= 0 \\ c_1 + c_3 &= 0 \end{aligned}$$

From equation 3: $c_2 = 0$

Substitute into equation 1: $2c_1 = 0 \Rightarrow c_1 = 0$

Substitute into equation 4: $c_3 = 0$

Conclusion: The **only** solution is $c_1 = c_2 = c_3 = 0$.

The vectors are **linearly independent**!

Key Insight: Independence = No Redundancy

Linear independence means:

- No recipe can be written as a combination of others
- Each recipe contributes something **new**
- **No wasted vectors** in the list

Linear dependence would mean:

- Some recipe is redundant (can be made from others)
- For example: If 🥛 = 0.5☕ + 0.5🍵, then v_3 is redundant

Left Cancellation Property

Change of Perspective

What we've learned so far:

- **What** linear independence means (physically: no redundant recipes)
- **How** to test it (check if $c_1 v_1 + \cdots + c_k v_k = 0 \Rightarrow$ all $c_i = 0$)

From This Point Forward

We **forget about coffee shop recipes** and work with **pure matrix algebra**:

- **Vectors** $v_1, \ldots, v_k$ become **matrix columns** $U = \begin{pmatrix} | & & | \\ v_1 & \cdots & v_k \\ | & & | \end{pmatrix}$
- **Linear independence** becomes a **cancellation property**

This simplifies our work: we only track **matrix operations**, not physical meanings.

Matrix Form of Linear Independence

Collect vectors $v_1, \ldots, v_k$ as columns of matrix U:

$$U = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_k \\ | & | & & | \end{pmatrix}$$

Linear independence says:

$$U \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} = 0 \quad \Rightarrow \quad \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} = 0$$

In other words: $Uc = 0 \Rightarrow c = 0$

Left Cancellation Property

Proposition (Left Cancellation)
If U has linearly independent columns, then:

$$UP = UQ \quad \Rightarrow \quad P = Q$$

Proof: If $UP = UQ$, then:

$$UP - UQ = 0$$

$$U(P - Q) = 0$$

Since U has linearly independent columns, $U(P - Q) = 0 \Rightarrow P - Q = 0$.

Therefore $P = Q$. $\square$

Why This Matters: Unique Representations

Left cancellation means **unique representation**:

If $v_1, \ldots, v_k$ are linearly independent and we write:

$$w = c_1 v_1 + \cdots + c_k v_k$$

then the coefficients $c_1, \ldots, c_k$ are **unique**.

Proof: If also $w = d_1 v_1 + \cdots + d_k v_k$, then:

$$U \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = U \begin{pmatrix} d_1 \\ \vdots \\ d_k \end{pmatrix}$$

By left cancellation: $c_i = d_i$ for all i. $\square$

Trace and Dimension

The Dimension Question

Definition (Basis)
A **basis** is a list of vectors that is:

- **Linearly independent**
- **Spans** the space (every vector can be written as a linear combination)

Question: If a space has two different bases, do they have the **same number** of vectors?

- Basis 1: $e_1, e_2, \ldots, e_n$
- Basis 2: $w_1, w_2, \ldots, w_m$

Is $n = m$?

The Tool: Trace of a Matrix

Definition (Trace)
The **trace** of a square matrix A is the sum of its diagonal entries:

$$\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

Example:

$$\operatorname{tr}\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = 1 + 5 + 9 = 15$$

Key property we need: $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ (even if A and B are **not square**!)

Visual Proof: $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ — First Diagonal Entry

Consider 2×3 matrix A and 3×2 matrix B:

$$\overset{A}{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}} \quad \overset{B}{\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}} \quad \overset{AB}{\begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}}$$

First diagonal entry (gray): sum of products of matching colors

$$c_{11} = {\color{orange}a_{11} b_{11}} + {\color{blue}a_{12} b_{21}} + {\color{green}a_{13} b_{31}}$$

Visual Proof: Second Diagonal Entry

For the second diagonal entry of AB:

$$\overset{A}{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}} \overset{B}{\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}} \overset{AB}{\begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}}$$

Second diagonal entry:

$$c_{22} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32}$$

Visual Proof: tr(AB) — Combined

Combining both diagonal entries:

$$\begin{aligned}\mathrm{tr}(AB) &= c_{11} + c_{22} \\ &= (a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31}) \\ &\quad + (a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32})\end{aligned}$$

This is the sum of all **color-matched pairs** from the two matrices.

Visual Proof: tr(BA) — Same Colors!

Now compute BA (note: B is 3×2, A is 2×3, so BA is 3×3):

$$\underbrace{\begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix}}_{B} \underbrace{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} d_{11} & \cdots & \cdots \\ \cdots & d_{22} & \cdots \\ \cdots & \cdots & d_{33} \end{pmatrix}}_{BA}$$

Diagonal entries of BA:

$$d_{11} = b_{11}a_{11} + b_{12}a_{21}$$
$$d_{22} = b_{21}a_{12} + b_{22}a_{22}$$
$$d_{33} = b_{31}a_{13} + b_{32}a_{23}$$

Visual Proof: Conclusion

$$\mathrm{tr}(BA) = d_{11} + d_{22} + d_{33}$$

$$= (b_{11}a_{11} + b_{12}a_{21}) + (b_{21}a_{12} + b_{22}a_{22}) + (b_{31}a_{13} + b_{32}a_{23})$$

By commutativity of multiplication ($a_{ij}b_{kl} = b_{kl}a_{ij}$):

$$= (a_{11}b_{11} + a_{21}b_{12}) + (a_{12}b_{21} + a_{22}b_{22}) + (a_{13}b_{31} + a_{23}b_{32})$$

This is **exactly the same sum** as $\mathrm{tr}(AB)$!

$$\boxed{\mathrm{tr}(AB) = \mathrm{tr}(BA)}$$

Key: The colored pairs are the same, just encountered in different order.

Using Trace to Prove Dimension is Well-Defined

Theorem
If V has two bases with n and m vectors respectively, then $n = m$.

Proof strategy:

1. Write one basis in terms of the other using change-of-basis matrices
2. Use left cancellation to show these matrices compose to identity
3. Use trace to count the sizes

Proof: Change of Basis Matrices

Let $e_1, \ldots, e_n$ be one basis and $w_1, \ldots, w_m$ be another basis.

Since $\{w_i\}$ is a basis, we can write each e_j as a linear combination:

$$\begin{pmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ w_1 & \cdots & w_m \\ | & & | \end{pmatrix} P$$

for some $m \times n$ matrix P.

Similarly, we can write:

$$\begin{pmatrix} | & & | \\ w_1 & \cdots & w_m \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{pmatrix} Q$$

for some $n \times m$ matrix Q.

Proof: Composing the Changes

Substituting the second equation into the first:

$$\begin{pmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ e_1 & \cdots & e_n \\ | & & | \end{pmatrix} QP$$

Since $\{e_i\}$ are linearly independent, by **left cancellation**:

$$QP = I_n$$

Similarly, substituting the first equation into the second:

$$\begin{pmatrix} | & & | \\ w_1 & \cdots & w_m \\ | & & | \end{pmatrix} = \begin{pmatrix} | & & | \\ w_1 & \cdots & w_m \\ | & & | \end{pmatrix} PQ$$

So $PQ = I_m$ by left cancellation.

Proof: Using Trace

We have:

- $QP = I_n$ (an $n \times n$ identity matrix)
- $PQ = I_m$ (an $m \times m$ identity matrix)

Taking traces:

$$\mathrm{tr}(QP) = \mathrm{tr}(I_n) = n$$

$$\mathrm{tr}(PQ) = \mathrm{tr}(I_m) = m$$

But by the trace property $\mathrm{tr}(QP) = \mathrm{tr}(PQ)$:

$$n = m$$

Therefore, **any two bases have the same number of vectors**! $\square$

Definition: Dimension

Definition (Dimension)
The **dimension** of a vector space V is the number of vectors in any basis of V.

Notation: $\dim(V)$

This is **well-defined** because we just proved all bases have the same size!

Examples:

- $\dim(\mathbb{R}^n) = n$ (basis: standard vectors $e_1, \ldots, e_n$)
- $\dim(\mathrm{Col}(A)) =$ number of vectors in any basis for the column space

Rank is Well-Defined

Connecting Rank to Dimension

Theorem
The columns of U from cross-filling $A = UV$ form a **basis** for $\mathrm{Col}(A)$.

Therefore:

$$\mathrm{rank}(A) = \text{number of columns of } U = \dim(\mathrm{Col}(A))$$

Since dimension is well-defined (doesn't depend on choice of basis), **rank is well-defined** (doesn't depend on pivot choices in cross-filling)!

Why U Forms a Basis — Preview

The proof requires two parts:

Part 1: Columns of U **span** $\text{Col}(A)$

Every column of A can be written as a linear combination of columns of U (from $A = UV$).

Part 2: Columns of U are **linearly independent**

This comes from the triangular structure created by the pivot selection in cross-filling.

(Full proof in next lecture — we need one more tool first)

Summary

What we proved today:

1. **Linear independence** = unique representation
2. **Left cancellation** property for independent columns
3. **Trace property**: $\text{tr}(AB) = \text{tr}(BA)$ (visual proof with colors)
4. **Dimension is well-defined**: All bases have the same size (using trace)
5. **Rank is well-defined**: $\text{rank}(A) = \dim(\text{Col}(A))$

Big picture: We elevated rank from an algorithmic definition to a **geometric property**.